

ASSESSMENT TOOL

SHORT ANSWERS

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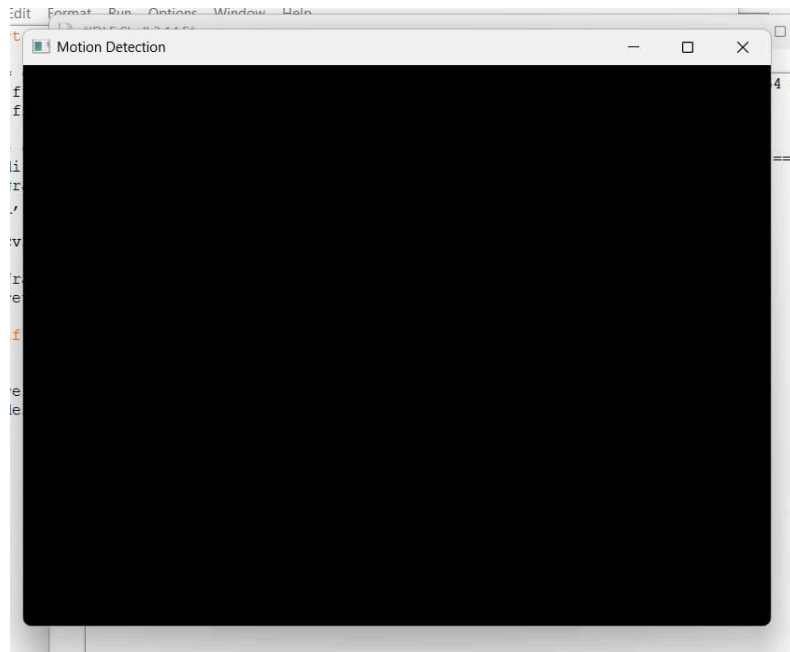
REG NO: 192521079

13. An automated surveillance system monitors moving objects. Write a Python

program for motion detection.

PROGRAM:

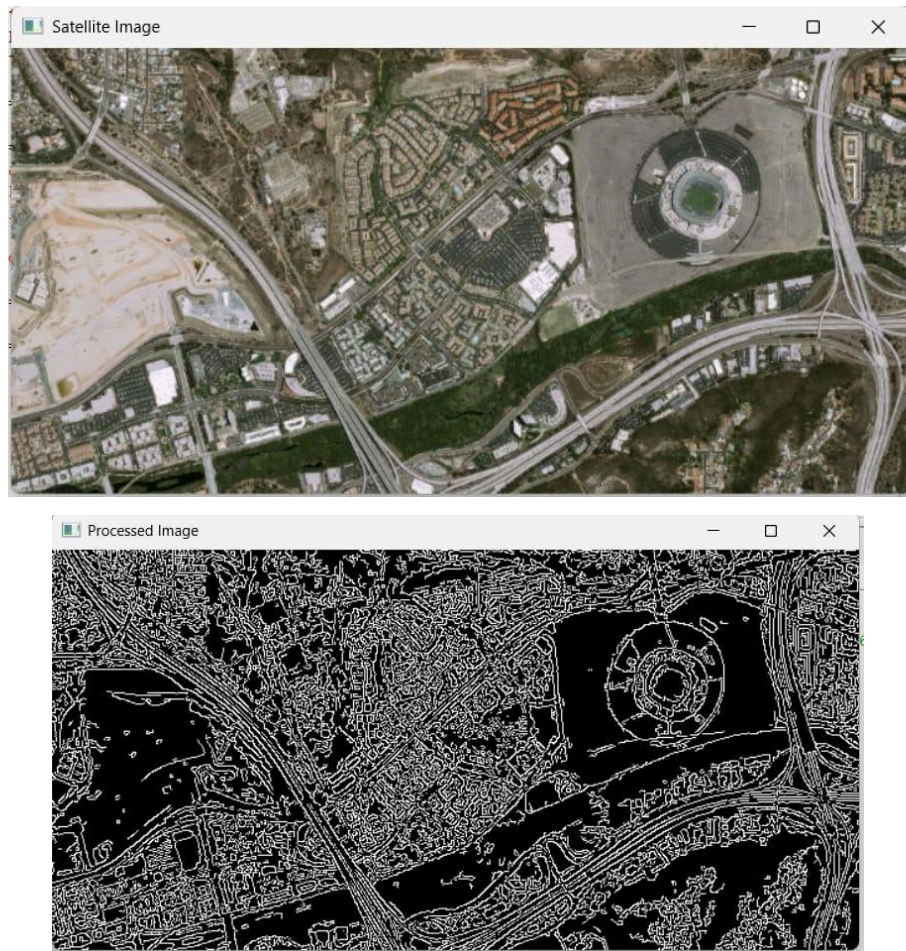
```
import cv2
cap = cv2.VideoCapture(0)
ret, frame1 = cap.read()
ret, frame2 = cap.read()
while cap.isOpened():
    diff = cv2.absdiff(frame1, frame2)
    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)
    _, thresh = cv2.threshold(gray, 25, 255, cv2.THRESH_BINARY)
    cv2.imshow("Motion Detection", thresh)
    frame1 = frame2
    ret, frame2 = cap.read()
    if cv2.waitKey(1) == 27: # Press ESC to exit
        break
cap.release()
cv2.destroyAllWindows()
```



14. A satellite image is used for land analysis. Write a Python program to process a satellite image.

PROGRAM:

```
import cv2  
img = cv2.imread("satellite.jpg")  
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)  
edges = cv2.Canny(gray, 100, 200)  
cv2.imshow("Satellite Image", img)  
cv2.imshow("Processed Image", edges)  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```



15. An OCR application processes text from images. Write a Python program for basic OCR.

PROGRAM:

```
import cv2
import pytesseract
img = cv2.imread("text_image.png")
text = pytesseract.image_to_string(img)
print("Extracted Text:")
print(text)
```

